Neuromechanics of Human-Exoskeleton Interaction: Brain, Body and Experience

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LOKOASSIST

Motivation and Research Challenge

The problem

- Exoskeletons are usually tuned to save **energy** or **muscle activity** [1]
- But people decide whether to keep a device by **how it feels**
- Which factors do people actually weigh when they judge assistance? **We do not know**.

Why it matters

- A device that saves energy but feels wrong will be rejected
- Knowing the user's own criteria allows assistance that people **accept and want to use** [2]

Scan for the Questionnaire

THE HUMAN COST FUNCTION

$$\text{Judgement} = \Sigma\, w_i \cdot \text{term}_i$$

$$w_i = ?$$

Which terms matter, and how much, for each person?

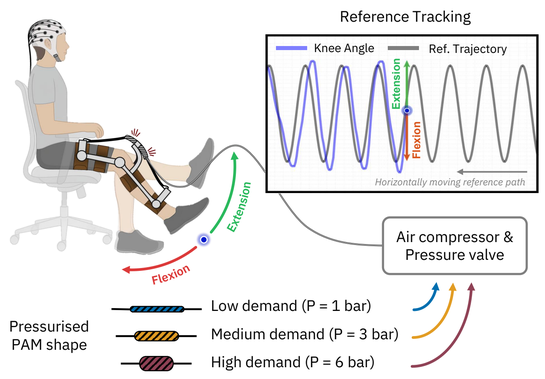
Energy | Muscle Effort | Stability | Naturalness | Predictability | Control

My Approach

Two studies: what makes a device feel right?

From a controlled single joint movement to **natural walking**

Study 1 · done: knee exoskeleton, seated



- 14 participants
- A visuo-motor tracking task while the exoskeleton applies 3 levels of physical demand
- EEG, knee flexors and extensors, and subjective evaluations were recorded

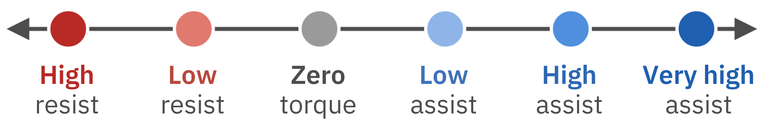
Study 2 · starting now: hip exoskeleton, walking

Brain: 64-channel mobile EEG

Body: EMG, Kinematics (IMU), GRF, energy cost

Experience: Custom-designed questionnaire

24 participants, 2 days each, 6 modes:





Link to the LokoAssist core ideas

- **Dual evaluation:** expert measures (EEG, muscles, energy, gait kinematics) linked to each user's own judgement
- **Dual functionality:** the exoskeleton assists, and its six modes probe how brain and body respond

How Could Others Help Me? How Could Others Benefit From my Research?

I am looking for...

- **Psychology and human factors:** validating short rating scales; agency, embodiment and trust

I can offer...

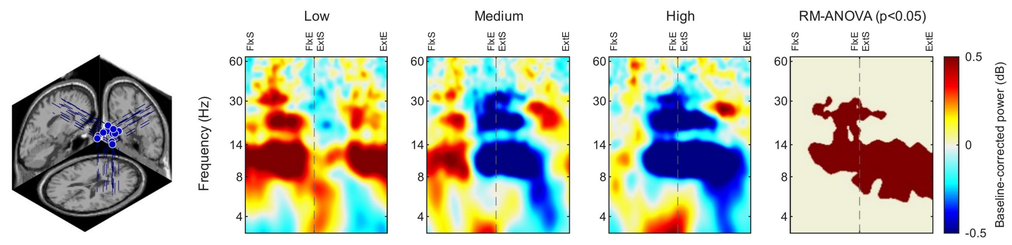
- **An open dataset:** brain, body and experience, different interaction levels
- **An open questionnaire** for your own device study
- **Our mobile brain/body lab:** add EEG and muscle recordings to your study

Current Status

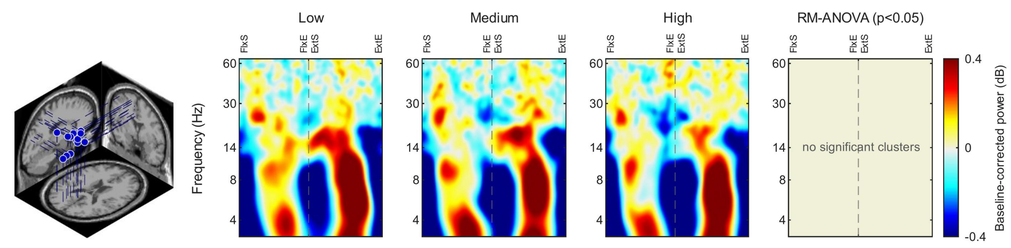
Study 1: knee exoskeleton · paper ready [3]

A unique regional fingerprint of the physical demand imposed by the exoskeleton on the cortical dynamics

Right sensorimotor cortex (n = 12):
Alpha & beta power decrease step by step with demand



Left parieto-occipital cortex (n = 13):
Task-locked modulation in theta & alpha power, independent of the demand



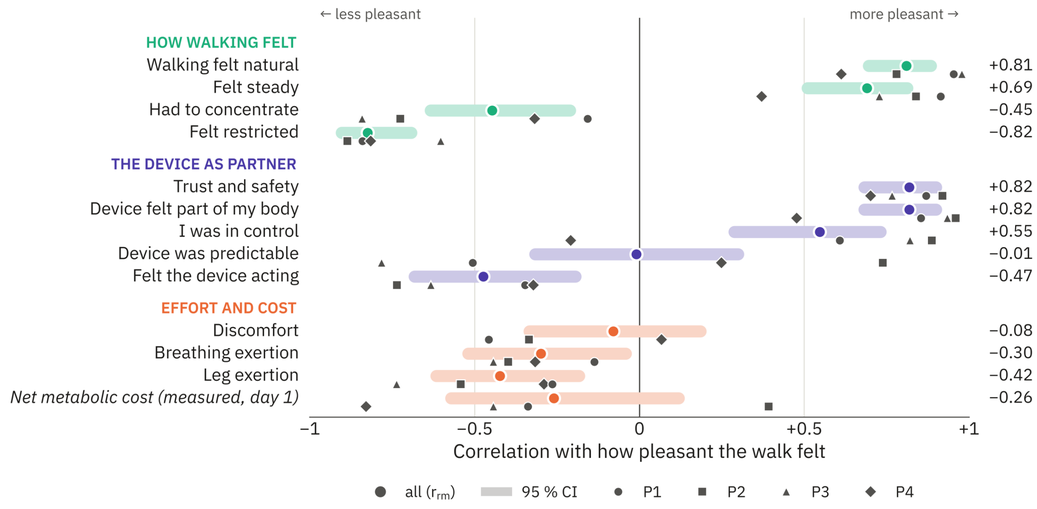
Behaviour: Perceived difficulty increased from 1.8 to 7.0 unit

Muscles 26% | Not explained by muscles or tracking error: 73%

Study 2: hip exoskeleton (pilot, n = 4)

- Compared with zero torque, resisting modes raised energy cost by about 10 to 25%; high assistance saved about 5 %
- The **strongest assistance did not feel like help** (3 of 4 rated it as hindering)

What goes with a pleasant walk? How walking felt, not energy



Repeated-measures correlation [4]; descriptive only

What remains unclear

- Which brain signals carry the **judgement**, not only the demand?

[1] Sawicki et al. (2020) J NeuroEng Rehabil 17:25. [2] Ingraham, Remy & Rouse (2022) Sci Robot 7:eabj3487. [3] Khosrotabar, Gramann, Seyfarth, Vogt & Sharbafi. Sensorimotor cortex reflects human-exoskeleton interaction level (manuscript). [4] Bakdash & Marusich (2017) Front Psychol 8:456.

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